

ISO/TC 22/SC 12 N

Date: 2013-04-12

ISO/PDTS 17242

ISO/TC 22/SC 12/WG 3

Secretariat: AFNOR

Road vehicles — Quasi-static Calibration Procedure for Belt Force Transducers

Véhicules routiers — Procédure d'étalonnage quasi-statique pour capteurs d'efforts pour ceintures

Warning

This document is not an ISO International Standard. It is distributed for review and comment. It is subject to change without notice and may not be referred to as an International Standard.

Recipients of this draft are invited to submit, with their comments, notification of any relevant patent rights of which they are aware and to provide supporting documentation.

Document type: Technical Specification

Document subtype:

Document stage: (20) Preparatory

Document language: E

C:\Users\gso\AppData\Local\Microsoft\Windows\Temporary Internet
Files\Content.Outlook\VQ35BR5M\D630_DRAFT_Test_Conditions_EN_Seat_Belt_Load_Cell_Calibration_Pr
ocedure_.doc STD Version 2.5a

Copyright notice

This ISO document is a working draft or committee draft and is copyright-protected by ISO. While the reproduction of working drafts or committee drafts in any form for use by participants in the ISO standards development process is permitted without prior permission from ISO, neither this document nor any extract from it may be reproduced, stored or transmitted in any form for any other purpose without prior written permission from ISO.

Requests for permission to reproduce this document for the purpose of selling it should be addressed as shown below or to ISO's member body in the country of the requester:

[Indicate the full address, telephone number, fax number, telex number, and electronic mail address, as appropriate, of the Copyright Manager of the ISO member body responsible for the secretariat of the TC or SC within the framework of which the working document has been prepared.]

Reproduction for sales purposes may be subject to royalty payments or a licensing agreement.

Violators may be prosecuted.

Contents

Page

Foreword	iv
Introduction.....	v
1 Scope	1
2 Field of application	1
3 Tests conditions	2
4 Calibration Procedure	3
Annex A (normative) Test setup	4
Annex B (normative) Evaluation method	5
Linear Regression	5
Polynomial Regression.....	7
Annex C (normative) Belt strap characteristics	8
Specifications for the Standardized Calibration Belt Strap	8
Measurement of Test Belt Strap Thickness	9
Annex D (non-normative) Applications Notes	10
Application of the Polynomial Scaling	10
Application of Unified Measurement Ranges	10
Application of Correct Linear Scaling	10

Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

International Standards are drafted in accordance with the rules given in the ISO/IEC Directives, Part 2.

The main task of technical committees is to prepare International Standards. Draft International Standards adopted by the technical committees are circulated to the member bodies for voting. Publication as an International Standard requires approval by at least 75 % of the member bodies casting a vote.

In other circumstances, particularly when there is an urgent market requirement for such documents, a technical committee may decide to publish other types of normative document:

- an ISO Publicly Available Specification (ISO/PAS) represents an agreement between technical experts in an ISO working group and is accepted for publication if it is approved by more than 50 % of the members of the parent committee casting a vote;
- an ISO Technical Specification (ISO/TS) represents an agreement between the members of a technical committee and is accepted for publication if it is approved by 2/3 of the members of the committee casting a vote.

An ISO/PAS or ISO/TS is reviewed after three years in order to decide whether it will be confirmed for a further three years, revised to become an International Standard, or withdrawn. If the ISO/PAS or ISO/TS is confirmed, it is reviewed again after a further three years, at which time it must either be transformed into an International Standard or be withdrawn.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights.

ISO/TS 17242 was prepared by Technical Committee ISO/TC 22, *Road vehicles*, Subcommittee SC 12, *Passive safety crash protection systems*.

This second/third/... edition cancels and replaces the first/second/... edition (), [clause(s) / subclause(s) / table(s) / figure(s) / annex(es)] of which [has / have] been technically revised.

Introduction

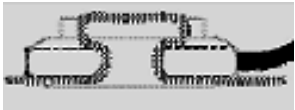
Road vehicles — Quasi-static Calibration Procedure for Belt Force Transducers

1 Scope

The objective of this document is to provide a procedure to calibrate seat belt force transducers with loading capacities up to 25kN and consistent test specifications and sequences in order to improve comparability of measurement results between testing laboratories.

2 Field of application

The described measurement procedure refers to belt force transducers with the specifications identified below and a loading capacity of up to 25kN.

Type	Description	Belt Orientation
1	baffle (test belt strap according to annex 3)	
2	loop (test belt strap according to customer requirement)	
3	clamping (test belt strap according to customer requirement)	
4	3 (test belt strap according to customer requirement)	bolt 

Limitations for the application of the test belt strap

In general, the standardized calibration belt strap (see annex 3.1) is only recommended for the calibration of belt force transducers which are to be utilized in tests that focus on comparability between different laboratories (such as during development efforts). **In case absolute values have to be obtained the seat belt webbing designated for the later testing purposes is to be used for calibration. Despite this the measurement of absolute values seems to be questionable as various investigations show significant variation. Otherwise a different measurement principle (e.g. another transducer type) must be utilized.**

The calibration by using the standardized calibration belt strap is only allowed for highly dynamic loaded transducers as shown in Type 1 (baffle). Consequently Type 2, 3, 4 and *Type 1 transducers for quasi-static loads and measurement ranges up to 500N* (i.e. comfort measurements) are to be excluded.

NOTE If required (e.g. by the user or the test protocol) *Type 1 transducers with measurement ranges beyond 500N* can be calibrated with a specimen of the seat belt webbing which is designated for later testing purposes. For the calibration of Type 2, 3, 4 and *Type 1 transducers for quasi-static loads and measurement ranges below 500N*, a specimen of the—seat belt webbing which is designated for the later testing purposes will be necessary. The belt strap type used for calibration as well as the strap's average thickness (see Annex 3) is to be identified in the calibration report.
If a non-standardized belt strap is given it is recommended to perform also a calibration with the standardized belt strap as reference in the scope of the belt force transducer calibration.

3 Tests conditions

3.1.0	Test method	Continuous loading up to transducer's full scale range ¹
3.2.0	Clamping length in mm	375 ± 75
3.3.0	Test velocity in mm/min	170 ± 30
3.4.0	Belt strap	Standardized belt strap (see specification in Annex 3) ²
3.5.0	Load relieving	Hysteresis not evaluated
3.6.0	Data Acquisition	Amplifier offset must be considered for the error of measurement
3.6.1	Filter settings	To be defined
3.6.2	Offset correction	No offset correction of test transducer prior calibration test
3.7.0	Evaluation	First loading ³
3.7.1	Evaluated range	From 2% of minimum transducer range to full scale range (e.g. 0.32kN-16kN @ 16kN sensor range) ⁴
3.7.2	Non-Linearity	Displayed as percentage in % of the measurement range
3.7.3	Sampling rate	Minimum of 100Hz ⁵
3.8.0	Excitation [V]	Voltage and sensor connection in accordance to customer requirements ⁶
3.9.0	Environmental conditions	Monitored
3.9.1	Temperature in °C	21 ± 2
3.9.2	Relative humidity in %	10 to 70

¹ See transducer data sheet

² Other belt strap types have to be identified in the calibration report

³ No preloading procedures

⁴ Ensure the comparison standard is traceable (e.g. ISO 376)

⁵ To reach appropriate accuracy

⁶ Excitation voltage as well as pin assignment and the use of Sense are to be stated in calibration report

4 Calibration Procedure

4.1.0 Conditioning

4.1.1 Allow adequate warm-up of data acquisition, test and reference transducer until thermal equilibrium

4.2.0 Test preparation

4.2.1 Parameters entry (serial number, temperature, humidity, belt strap type etc.)

4.2.2 Measure sensor offset in unloaded condition and record the value in mV/V in the calibration report

4.2.3 Insert new belt strap⁷, ensure centered position of the belt strap in the fixations

4.2.4 Centered and perpendicular mounting of test transducer (see instructions in user's manual)

4.2.5 The belt strap preload must not exceed 0.2% of the transducer's full scale range (while holding the transducer in place)

4.3.0 Calibration test

4.3.1 Do not zero the transducer offset⁸

4.3.2 One loading sequence with one belt strap per test transducer

4.3.3 Up to a force of 101% of the transducer's full scale range the test is to be performed within the given velocity ranges (see Test Conditions); the calibration data is to be acquired continuously

4.3.4 There are no specifications for the load relieving

4.4.0 Data storage

4.4.1 Determine a Third Degree Polynomial (see Annex 2) from the recorded data points; the data storage format is optional⁹

4.5.0 Data evaluation

4.5.1 Evaluation in accordance to Annex 2¹⁰:

Linear Regression: least-squares fit method without any offset correction

Polynomial Regression: least-squares fit method without any offset correction

4.6.0 Documentation

4.6.1 Create a test report and record temperature, humidity, belt strap type, strap thickness¹¹, sensitivity in kN/mV/V and mV/kN/V⁶ as well as the intercept D¹², non-linearity as a percentage in % of the transducer's full scale range, the third degree polynomial equation in the form $F=AS^3+BS^2+CS+M^{13}$, polynomial coefficients in kN/mV/V, polynomial coefficient's significant digits: 3rd degree: 9 (e.g. 0,00123456789), 2nd degree: 6 (e.g. 1,23456), 1st degree: 3 (e.g. 12,3)

⁷ Fixation by clamp or loop (ECE-R16 conform)

⁸ Use of the calibration coefficient as defined in SAE-J2517: Prior a crash test, the original zero offset level must be preserved by either not zeroing the transducer or the amount that was zeroed must be added during post-processing.

⁹ ISO 13499 recommended: incl. time channel, output test sensor, output reference sensor, and travel distance

¹⁰ If the calibrated transducer is used in tests the hardware offset must not be corrected

¹¹ See page 13 "Measurement of Test Belt Strap Thickness"

¹² Calculated offset from the linear regression (see page Linear Regression⁹ "Linear Regression")

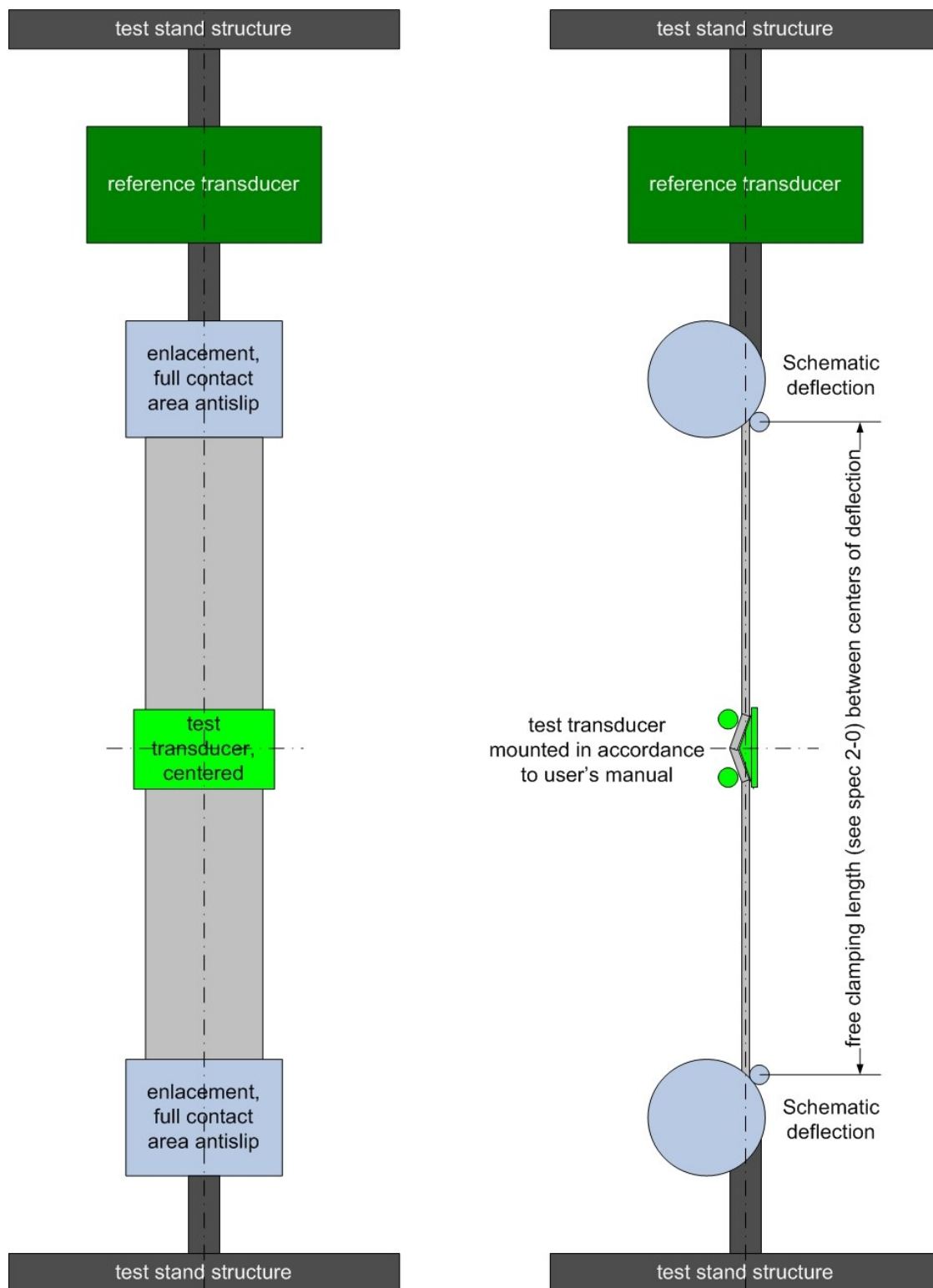
¹³ See page 11 "Polynomial Regression"

and the measured sensor offset in unloaded condition in mV/V¹⁴

¹⁴ To check the offset magnitude of the data acquisition system

Annex A (normative)

Test setup



Annex B (normative)

Evaluation method

Linear Regression

For the software linearization, use a least-squares fit method:

This measurement is independent of the defined evaluation interval (Test Condition 2.7.1). The resulting linear equation is:

$$F_i = bS_i + D$$

For the linearization, the square sum of the difference between measured value and linearized value must be minimal:

$$\sigma = \sum_{i=1}^n [D + bS_i - F_i]^2$$

$$S_i, F_i = \text{measured}$$

The differentiation is equated to zero:

$$\frac{d\sigma}{db} = 2 \sum_{i=1}^n [S_i D + S_i b - S_i F_i] = 0$$

$$\frac{d\sigma}{dD} = 2 \sum_{i=1}^n [D + S_i b - F_i] = 0$$

From this will follow:

$$\sum_{i=1}^n [D + S_i b] = \sum_{i=1}^n [F_i]$$

$$\sum_{i=1}^n [S_i D + S_i b] = \sum_{i=1}^n [S_i F_i]$$

This results in a matrix form:

$$\begin{bmatrix} n & \sum_{i=1}^n S_i \\ \sum_{i=1}^n S_i & \sum_{i=1}^n S_i^2 \end{bmatrix} \begin{pmatrix} D \\ b \end{pmatrix} = \begin{bmatrix} \sum_{i=1}^n F_i \\ \sum_{i=1}^n F_i S_i \end{bmatrix}$$

$$\begin{bmatrix} n & \sum_{i=1}^n S_i \\ \sum_{i=1}^n S_i & \sum_{i=1}^n S_i^2 \end{bmatrix} = A$$

And can be solved by multiplying with the inverted matrix A:

$$A^T \begin{bmatrix} n & \sum_{i=1}^n S_i \\ \sum_{i=1}^n S_i & \sum_{i=1}^n S_i^2 \end{bmatrix} \begin{pmatrix} D \\ b \end{pmatrix} = A^T \begin{bmatrix} \sum_{i=1}^n F_i \\ \sum_{i=1}^n F_i S_i \end{bmatrix}$$

$$A^T = \begin{bmatrix} \sum_{i=1}^n S_i^2 & -\sum_{i=1}^n S_i \\ -\sum_{i=1}^n S_i & n \end{bmatrix} \frac{1}{\det A}$$

$$\det A = n \sum_{i=1}^n S_i^2 - \left(\sum_{i=1}^n S_i \right)^2$$

$$\begin{pmatrix} D \\ b \end{pmatrix} = \frac{\begin{bmatrix} \sum_{i=1}^n S_i^2 & -\sum_{i=1}^n S_i \\ -\sum_{i=1}^n S_i & n \end{bmatrix} \begin{bmatrix} \sum_{i=1}^n F_i \\ \sum_{i=1}^n F_i S_i \end{bmatrix}}{\det A}$$

$$D = \frac{\sum_{i=1}^n S_i^2 \sum_{i=1}^n F_i - \sum_{i=1}^n S_i \sum_{i=1}^n F_i S_i}{\det A}$$

$$b = \frac{-\sum_{i=1}^n S_i \sum_{i=1}^n F_i + n \sum_{i=1}^n F_i S_i}{\det A}$$

$$a = \frac{1}{b}$$

F is the actual force in kN

S is the sensor output reading in mV/V

D is the physical offset prior to the test in kN (linear regression)

a is the sensitivity in mV/kN/V

b is the inverse sensitivity in kN/mV/V

σ is the deviation

Polynomial Regression

For the polynomial approximation, utilize a least-squares fit method to determine the polynomial coefficients A, B, C and M:

With mesh points $[S_i; F_i]$ the third degree polynomial will acquire the general form:

$$F_i = A \cdot S_i^3 + B \cdot S_i^2 + C \cdot S_i + M$$

The function values F_i are complemented by residues r_i (corrections):

$$F_i + r_i = A \cdot S_i^3 + B \cdot S_i^2 + C \cdot S_i + M$$

Applied on the mesh points the observation equations follow:

$$\begin{aligned} r_1 &= A \cdot S_1^3 + B \cdot S_1^2 + C \cdot S_1 + M - F_1 \\ \downarrow \\ r_i &= A \cdot S_i^3 + B \cdot S_i^2 + C \cdot S_i + M - F_i \end{aligned}$$

The observation equations are squared and then pooled:

$$\begin{aligned} \sum r_i^2 &= \sum A^2 \cdot S_i^6 \\ &+ 2 \sum AB \cdot S_i^5 + 2 \sum AC \cdot S_i^4 + 2 \sum AM \cdot S_i^3 \\ &- 2 \sum A \cdot F_i \cdot S_i^3 \\ &+ \sum B^2 \cdot S_i^4 \\ &+ 2 \sum BC \cdot S_i^3 + 2 \sum BM \cdot S_i^2 \\ &- 2 \sum B \cdot F_i \cdot S_i^2 + \sum C^2 \cdot S_i^2 + 2 \sum CM \cdot S_i - 2 \sum C \cdot F_i \cdot S_i + M^2 - 2M \cdot F_i + F_i^2 \end{aligned}$$

To determine the minimum of the residues r_i the observation equation's sum of squares are differentiated and equated to zero.

$$\begin{aligned} \frac{d(\sum r_i^2)}{dA} &= 2 \sum A \cdot S_i^6 + \sum B \cdot S_i^5 + \sum C \cdot S_i^4 + \sum M \cdot S_i^3 - \sum F_i \cdot S_i^3 = 0 \\ \frac{d(\sum r_i^2)}{dB} &= \sum A \cdot S_i^5 + 2 \sum B \cdot S_i^4 + \sum C \cdot S_i^3 + \sum M \cdot S_i^2 - \sum F_i \cdot S_i^2 = 0 \\ \frac{d(\sum r_i^2)}{dC} &= \sum A \cdot S_i^4 + \sum B \cdot S_i^3 + 2 \sum C \cdot S_i^2 + \sum M \cdot S_i - \sum F_i \cdot S_i = 0 \\ \frac{d(\sum r_i^2)}{dM} &= \sum A \cdot S_i^3 + \sum B \cdot S_i^2 + \sum C \cdot S_i + 2 \sum M - \sum F_i = 0 \end{aligned}$$

A linear system of equations results from putting in the mesh points $[x_i; y_i]$. Coefficients A, B, C and M and therefore the equalization polynomial through the point of origin can be calculated.

F is the actual force in kN

S is the sensor output reading in mV/V

M is the physical offset prior to the test in kN (polynomial regression)

A, B, and C are the calibration coefficients

Annex C (normative)

Belt strap characteristics

Specifications for the Standardized Calibration Belt Strap

For the calibration of the belt force transducers - only test belt straps with the specifications shown below should be utilized:

A3.1.1 Construction	Normalband, equal on both sides, bonding K2/2, doubleshot
A3.1.2 Width	47-49mm
A3.1.3 Thickness	1.25-1.35mm
A3.1.4 Weight	64-68g/m
A3.1.5 Min. breaking load	$\geq 30\text{kN}$
A3.1.6 Strain	14-16% bei 10kN
A3.1.7 Coloring / finishing	spin dying
A3.1.8 Number of chaining threads	300 ± 10
A3.1.9 Number of filling threads	66 ± 5 (1/100mm)

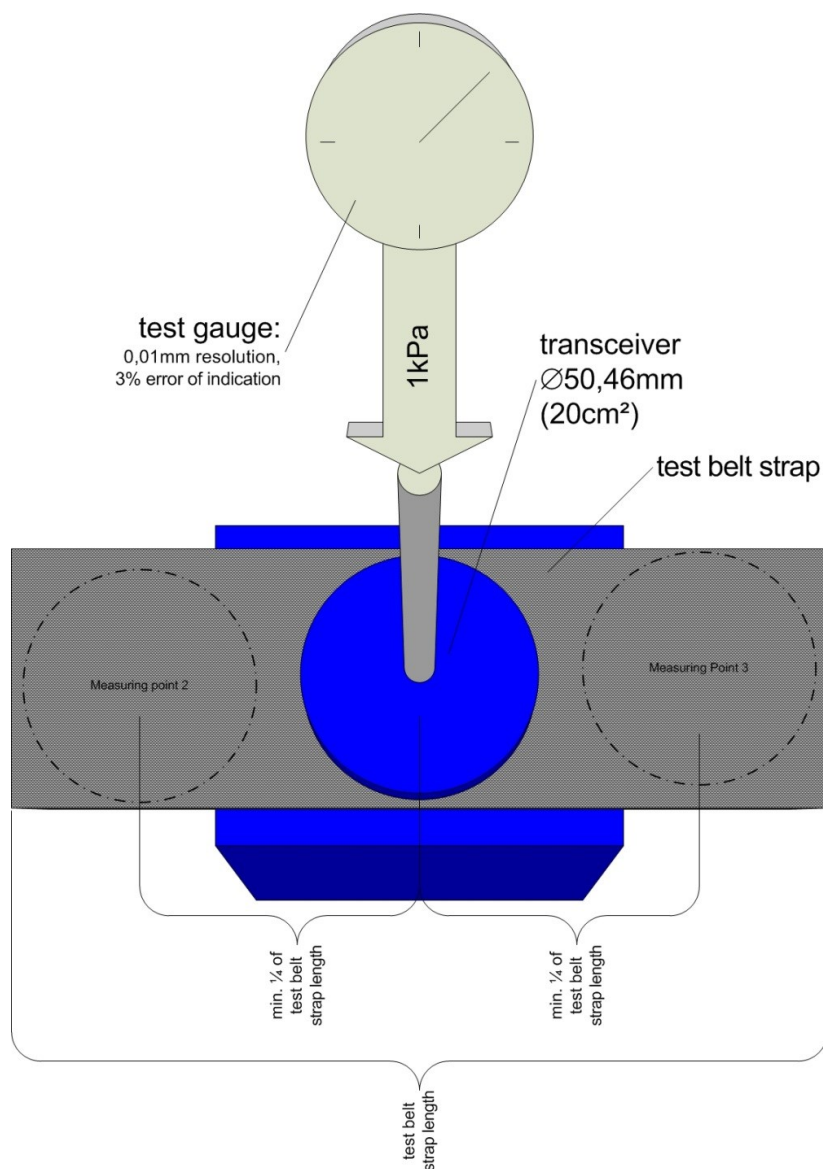
The belt strap has to be stored in a packed condition. Furthermore it should be protected from sunlight.

NOTE: The test belt strap as defined above is only permitted to be used for highly dynamic loaded transducers according to Type 1 (baffle). For other transducers, the test belt strap which is in accordance with customer requirements should be chosen.

Measurement of Test Belt Strap Thickness

For checking the significant belt strap characteristics, and to make sure that traceability is possible, the averaged thickness of the utilized belt strap charge has to be identified. The average and the classification of the belt strap must be noted in the calibration-report. Measurements of the test belt strap thickness are taken as follows¹⁵:

The test gauge should have a resolution of 0.01mm and a maximum indication error of 3%. The belt strap has to be measured between a planar rigid underlay and a flat transceiver with a surface area of 20cm² (caliber of ca. 50,46mm) within a test load from 1kPa. The averaged belt strap thickness is calculated by three measurements on different points in the medium range of the belt strap. The minimum distance between these points should be 25% of the overall tested belt strap length. If a belt strap charge is longer than 10m, the averaged belt strap thickness should be identified once per each 10m section.



¹⁵ Following DIN EN ISO 5084

Annex D (non-normative)

Applications Notes

Application of the Polynomial Scaling

By means of linearization electronics the non-linearity of belt force transducers can be reduced to less than 1%. In that case the higher accuracy of a polynomial regression fitting the calibration curve has no significant effect on the measurement results considering the error of measurement in the calibration institutes and in the crash test laboratories applying the polynomial scaling.

Therefore for process simplification the polynomial scaling is only recommended for belt force transducers without linearization electronics.

Application of Unified Measurement Ranges

To reach more accuracy with the linear regression over a certain measurement range below the transducer's full range a limitation of the calibrated range can be effective. The sensor's non-linearity leads to different transducer's sensitivities over different measurement ranges. That is to be considered for the comparability of the measurement results between laboratories applying different sensors calibrated in different ranges. **The recommended separation of measurement ranges of highly dynamic loaded belt force transducers is 6kN, 10kN and 16kN.**

To improve comparability laboratories should agree in utilizing transducer's of the same measurement range in the same test scenarios.

Application of Correct Linear Scaling

The linear regression delivers a linear equation in the form $F_i = bS_i + D$. D is the sensor offset which must be added to the value scaled by the sensitivity b to get the true measurement value. That must be considered for the linear scaling by software.